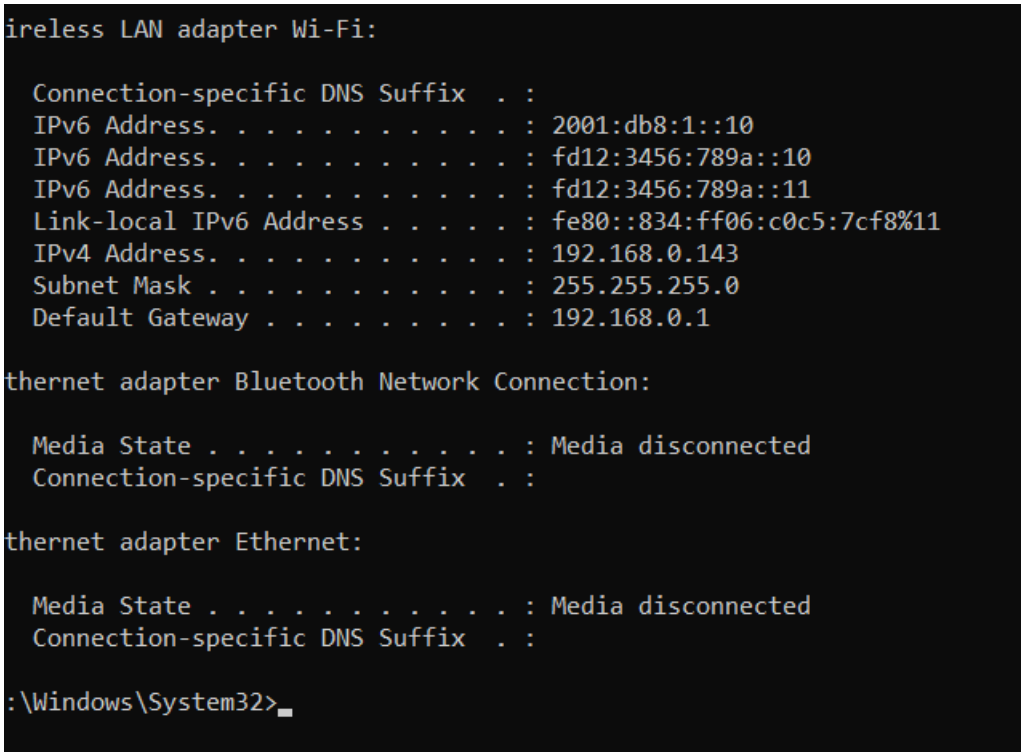
Task 1:

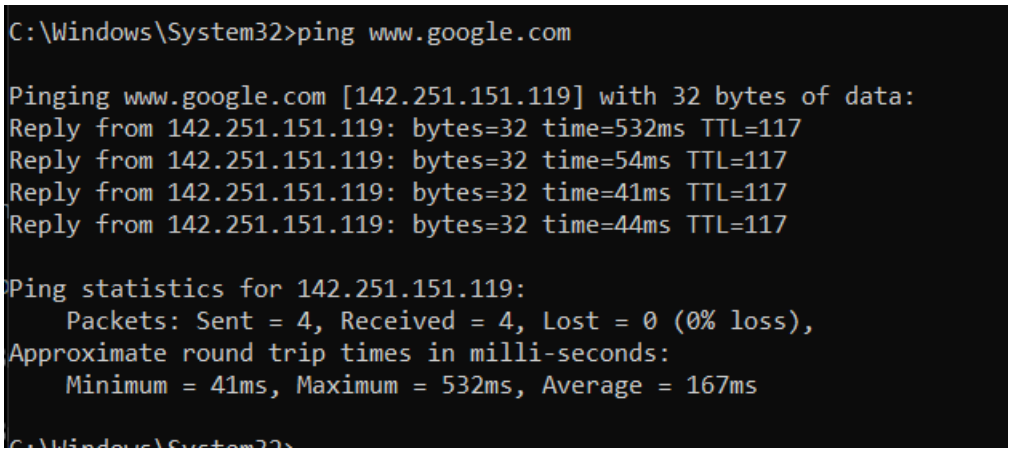
Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.



Task 2:

Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

Answer:



Task 3:

Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

Answer:

```
C:\Windows\System32>tracert www.flipkart.com

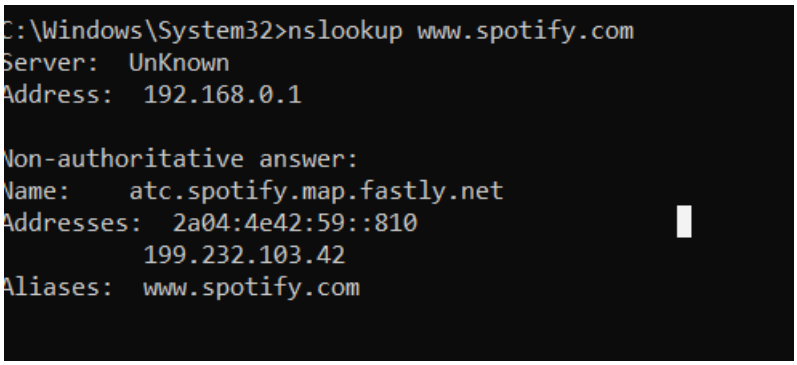
Tracing route to e348826.dsca.akamaiedge.net [23.223.245.44]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  192.168.0.1
  2     2 ms     2 ms     3 ms  Unit [192.168.1.1]
  3    18 ms    38 ms    33 ms  10.240.170.222
  4     *        *        *     Request timed out.
  5    32 ms    50 ms    32 ms  182.78.248.109
  6    68 ms    82 ms    76 ms  116.119.109.250
  7    69 ms   102 ms    68 ms  ae4.r03.border01.ccu01.fab.netarch.akamai.com [2.16.180.241]
  8    62 ms    78 ms    65 ms  lo1.r02.stem01.ccu01.fab.netarch.akamai.com [2.16.180.25]
  9    69 ms    93 ms    74 ms  lo1.r01.spine02.ccu01.fab.netarch.akamai.com [2.16.180.12]
 10    73 ms    67 ms   575 ms  lo1.r02.leaf01.ccu01.fab.netarch.akamai.com [2.16.180.5]
 11    64 ms    81 ms    83 ms  vlan100.r06.tor01.ccu01.fab.netarch.akamai.com [23.216.135.33]
 12    72 ms    67 ms    62 ms  a23-223-245-44.deploy.static.akamaitechnologies.com [23.223.245.44]

Trace complete.
```

Task 4:

Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.



The nslookup command checks DNS resolution by finding the IP address associated with a domain name.

Task 5:

A friend says they cannot access Zomato’s website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Answer:

Command	What it checks
ipconfig	Checks the computer's IP address, subnet mask, and default gateway to verify its network configuration.
ping www.zomato.com	Checks connectivity to Zomato's resolved IP address and measures response time, although some servers may block ping requests.
nslookup www.zomato.com	Checks whether DNS can resolve Zomato's domain name into an IP address.

Conclusion: These commands help identify whether the problem is caused by incorrect network settings, connectivity issues, or DNS resolution failure.